

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8521**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	401	Seat No:	2022600059
		Course Section:	1		

L_29_516_986_L_1505_38320



Name of the candidate. Pratham Shirbhate

Candidate's UID number:

2	0	2	2	5	0	0	0	5	9
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Examination class room number:

4	0	1
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Invigilator's signature with date:

<u>08/10/25</u>

(To be filled in by the candidate)

EXAMINATION: MSE - BE - BTECH
(For example FY MCA/FY BTECH)

SEMESTER: 7

PROGRAM: CSE - AIML

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: 08/10/25 - Wednesday

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Question
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Q1]

(b)

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Q2]

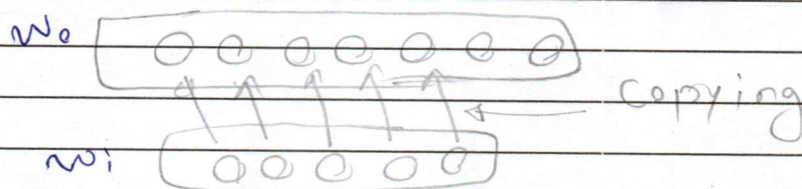
(b) (i) Overcomplete encoder is an encoder with hidden layer greater than input layer.

(ii) Which makes it just copy & paste input layer.

(iii) This creates the same copy hence it behaves like an identity function.

(iv)

$$\Rightarrow W_i = W_o$$



(v) To prevent this, we can regularize it with L2-term.

(vi) which will add a new λL term to objective function & other parameters.

(vii) Another trick is to tie its weights i.e. $W = W^T$

$$W_i = W$$

$$W_i^T = W^T$$

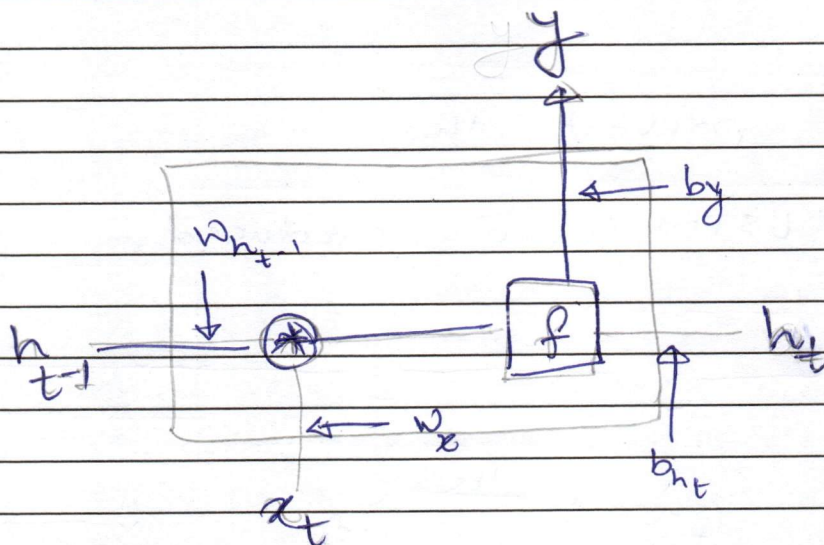
(viii) That is, train the model such that in input layer if w weight is used let the output layer weight be w^T .

(ix) This reduces complexity & easier to find the lowest reconstruction error.

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Q3)
(a)



$$h_t = f(w_x x_t + w_{h_{t-1}} \cdot h_{t-1} + b_{h_t})$$

$$\Rightarrow h_t = \sigma(w_x x_t + w_{h_{t-1}} \cdot h_{t-1} + b_{h_t})$$

$$y = \text{Softmax}(w_y x_t)$$

$$y = \text{Softmax}(w_y x_t + w_{y_{h_{t-1}}} \cdot h_{t-1} + b_y)$$

Gradient expression in sequencing of RNN

$$\Rightarrow \frac{\partial L}{\partial h_{t+k}} = \frac{\partial L}{\partial h_0} \left(\frac{\partial h_0}{\partial h_1} \right) \left(\frac{\partial h_1}{\partial h_2} \right) \dots \frac{\partial h_{t+k}}{\partial h_t}$$

$$= \frac{\partial L}{\partial h_0} \underbrace{\left(\frac{\partial h_0}{\partial h_1} \right) \left(\frac{\partial h_1}{\partial h_2} \right) \dots \frac{\partial h_{t+k}}{\partial h_t}}_{\text{Vanishing Gradient Problem}}$$

(Now, this part of the expression starts getting occupied by smaller numbers (ie. no). Hence, multiplication of numbers close to zero will be approx. zero.

$\therefore$ This thing is called vanishing gradient problem in RNNs in a long sentence or sequence.

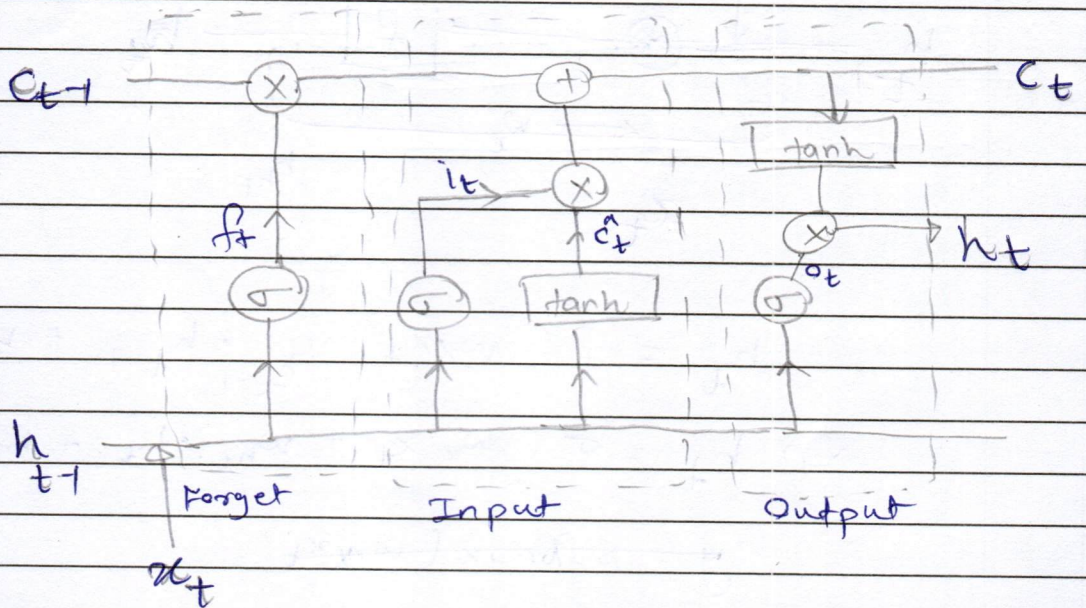
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To prevent this,

LSTM was introduced.



$$f_t = \sigma(w_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(w_i[h_{t-1}, x_t] + b_i)$$

$$\hat{c}_t = \tanh(w_c[h_{t-1}, x_t] + b_c)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \hat{c}_t$$

$$o_t = \sigma(w_o[h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(c_t)$$

(i) This new concept of forget gate decides which memory to forget & hence makes space for new memory to fit in.

(ii) The cell state is the memory state of the LSTM



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(iii) The cell state takes input of both forget gate & input gate with candidate cell hidden state.

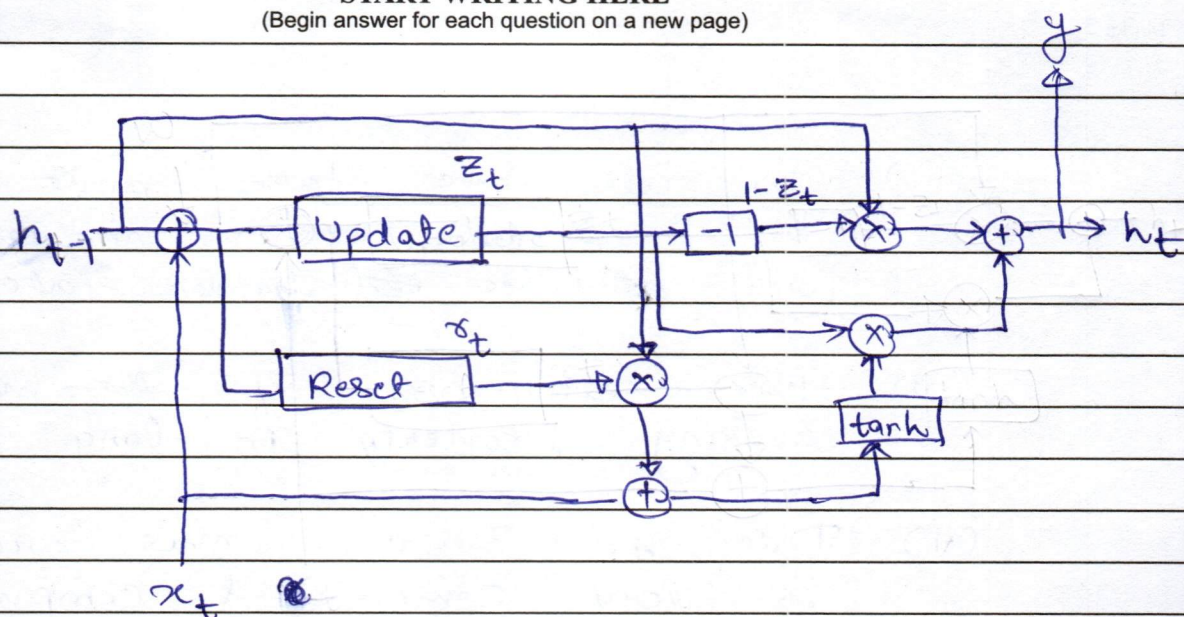
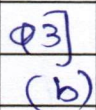
(iv) This summation keeps the memory working properly in long sentences.

(v) Due to 3 gates, this architecture is very complex & computationally heavy to train.

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(update gate) $z_t = \sigma(w_z x_t + w_{z h} \cdot h_{t-1} + b_z)$

(Reset gate) $\tau_t = \sigma(w_{\tau} \cdot x_t + w_{\tau h} \cdot h_{t-1} + b_{\tau})$

~~$$h_t = x_t \odot h_{t-1} + \tanh w$$~~

$$h_t = \left[\left[\tanh(x_t \odot h_{t-1} + w_h x_t) \odot [z_t] \right] + \left[[1 - z_t] \odot [h_{t-1}] \right] \right]$$

- (i) The update gate of GRU is a combination of both current input as well as previous output with the respective weights & biases.
- (ii) To compare GRUs with LSTMs, GRUs have less parameters (i.e. update & reset gate) while LSTM has 3 gates (input, output, forget).

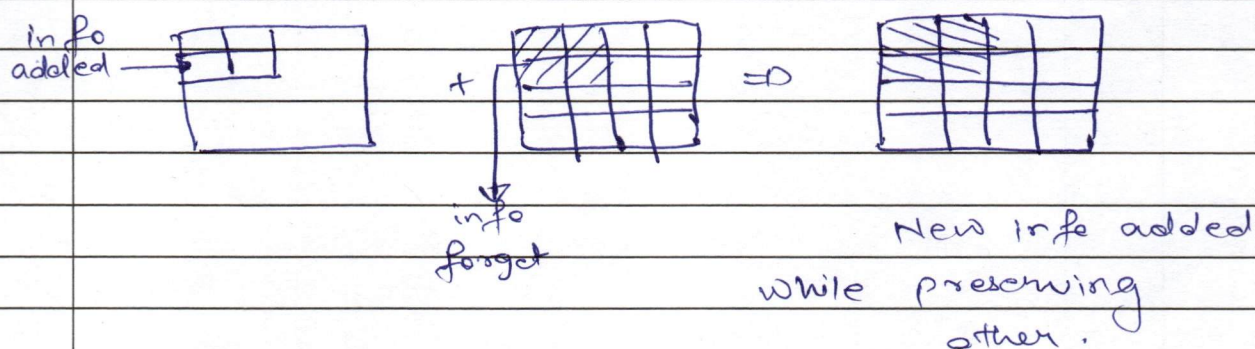
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(iii) This makes the GRU to be less computationally heavy than LSTM.

(iv) Because of this architecture, it's light weight & easier to scale.

(v) It works in a like give visualization:-



(vi) While LSTM fails to generalize at a very large memory, GRU can do it easily with this 2 gate architecture.

(vii) In real world applications, GRU have the capability to work with diff types of ~~ext~~ memory like a LLM or AI agent.

(viii) While LSTM, due to its memory cell state it can work better on specific type of memory like stock prediction.

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